Energy, Work & Power

Section 1: Work Done by Constant & Inclined Forces

1 A force of 60 N pulls a box across a horizontal floor through a distance of 8 m. The force acts at an angle of 60° above the horizontal. Find the work done by the force.

a. 240 J

b. 480 J

c. 415.7 J

d. 120 J

2 A crate of mass 15 kg is lifted vertically at a constant speed through a height of 4 m. Taking $g = 10\ \text{m/s}^2$, what is the work done against gravity?

a. 600 J

b. 150 J

c. 300 J

d. 60 J

3 A box is pushed 5 m along a horizontal floor against a constant frictional resistance of 35 N. What is the work done against friction?

a. 175 J

b. 150 J

c. 350 J

d. 70 J

Section 2: Kinetic Energy & Potential Energy

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4 Calculate the kinetic energy of an athlete of mass 70 kg running at a speed of 8 m/s.

a. 2240 J

b. 4480 J

c. 560 J

d. 280 J

5 A rollercoaster car of mass 500 kg starts from rest at the top of a frictionless hill of height 20 m. Taking $g = 10\ \text{m/s}^2$, find its speed at the bottom of the hill.

a. 20 m/s

b. 10 m/s

c. 40 m/s

d. 14.1 m/s

6 A bullet of mass 0.02 kg traveling at 300 m/s penetrates a wooden block to a depth of 0.15 m before stopping. What is the average resistive force exerted by the wood?

a. 6000 N

b. 3000 N

c. 9000 N

d. 12000 N

Section 3: The Work-Energy Principle

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7 A skier of mass 60 kg starts from rest at the top of a slope 30 m high. When reaching the bottom, her speed is 20 m/s. Taking $g = 10\ \text{m/s}^2$, how much energy was lost due to friction?

a. 6000 J

b. 12000 J

c. 18000 J

d. 3000 J

8 A package of mass 4 kg slides down a rough incline of length 10 m and vertical drop 6 m. If the resistive force is a constant 8 N, find the speed of the package at the bottom (starting from rest, $g = 10\ \text{m/s}^2$).

a. $\sqrt{80} \approx 8.94\ \text{m/s}$

b. 10 m/s

c. 8 m/s

d. 12 m/s

9 A particle is projected up a rough incline (30°) with initial speed 10 m/s. If $\mu = 0.2$ and $g = 10\ \text{m/s}^2$, how far up the plane does the particle travel before coming to rest?

a. ≈ 7.43 m

b. 10.0 m

c. 5.50 m

d. 8.25 m

Section 4: Power and Vehicles in Motion

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10 A car of mass 1200 kg travels along a level road at a constant speed of 30 m/s. The engine is working at 45 kW. What is the total resistance to motion?

a. 1500 N

b. 1200 N

c. 1350 N

d. 2000 N

11 The same car (mass 1200 kg, engine working at 45 kW) faces the same resistance of 1500 N while traveling at 20 m/s. What is its acceleration at this instant?

a. $0.625\ \text{m/s}^2$

b. $0.75\ \text{m/s}^2$

c. $1.00\ \text{m/s}^2$

d. $0.50\ \text{m/s}^2$

12 A truck of mass 5000 kg ascends an incline of 1 in 20 ($\sin\alpha = 1/20$) at a steady speed of 15 m/s. The engine power is 60 kW. Taking $g = 10\ \text{m/s}^2$, find the resistance to motion.

a. 1500 N

b. 2500 N

c. 4000 N

d. 2000 N

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